

Foot MRI Dictation Cheat Sheet - Expanded Findings & Impressions

Copy-ready language. Replace bracketed text. Use as a workstation reference, not a substitute for surgical correlation or local report style.

Search Pattern

- Forefoot pain: plantar plate, neuroma/bursa, stress injury, MTP cartilage.
- Midfoot pain: Lisfranc complex, navicular stress fracture, coalition, midfoot OA.
- Lisfranc: inspect dorsal/interosseous/plantar components plus alignment and marrow edema.
- Sesamoid pain: fracture vs bipartite, AVN, sesamoiditis.
- Report exact ray/space because surgical planning depends on it.

How to Find It — Landmark-Based Identification

Forefoot and midfoot MRI is read structure-by-structure off a few constant bony landmarks. Master the short-axis-to-the-metatarsals plane first, anchor yourself at the **second tarsometatarsal (TMT) joint**, and the rest of the anatomy hangs predictably off it.

Planes and Prescription

PEARL Prescribe short axis to the metatarsals

Forefoot studies are usually planned with the "**short axis**" prescribed perpendicular to the long axis of the **metatarsal shafts** (a true short axis to the rays), plus a **long-axis** plane along the shafts and a **coronal/sagittal** of the foot.

- The plane radiologists call **axial** in the forefoot is really **short-axis to the metatarsals** — it shows the rays in cross-section and is the workhorse for plantar plates, intermetatarsal spaces, and Lisfranc.
- **Fat-suppressed fluid-sensitive** (STIR or FS T2/PD) finds marrow edema and fluid; **T1** shows marrow fat and cortex; small FOV and thin slices are essential in this small anatomy.

LANDMARK The second TMT recess is your anchor

The **base of the second metatarsal** is recessed proximally between the **medial and lateral cuneiforms**, forming a mortise. This keystone is the origin point for the Lisfranc complex and the reference for normal TMT alignment.

Lisfranc Complex and TMT Alignment

HOW TO FIND IT Lisfranc ligament (interosseous component)

On **short-axis/axial**, find the strongest **interosseous Lisfranc ligament** running obliquely from the lateral aspect of the **medial cuneiform (C1)** to the **base of the second metatarsal (M2)**.

- There are three bands at this interval — **dorsal**, **interosseous (strongest)**, and **plantar**; the interosseous band is the key stabilizer.
- Normal: a thick **low-signal** band; trace it on consecutive short-axis slices from dorsal to plantar.

LANDMARK Normal TMT alignment landmarks

Check these constant alignments on axial/short-axis and correlate with radiographs:

- The **medial border of M2** lines up with the **medial border of the middle cuneiform (C2)**.
- The **medial border of M4** lines up with the **medial border of the cuboid**.
- Widening of the **C1–M2 interval** (diastasis) or a **fleck sign** (avulsion off M2 base) signals a Lisfranc injury.

PITFALL Plantar band is the prognostic one

The **plantar Lisfranc ligament** (C1 to bases of M2 and M3) is the most important for stability/outcome. Don't stop at the dorsal band — a torn or high-signal plantar component predicts instability even when alignment looks near-normal on static images.

PEARL Read Lisfranc as a complex, not a single ligament

Assess **dorsal**, **interosseous**, and **plantar** bands plus alignment and marrow edema at the M2 base and cuneiforms together. Bone marrow edema at the **second metatarsal base** is a sensitive red flag for subtle Lisfranc injury.

Lesser MTP Joints — Plantar Plate and Collaterals

HOW TO FIND IT Plantar plate of the lesser MTP joints (especially the 2nd)

The **plantar plate** is the fibrocartilaginous floor of the MTP joint. It is best seen on **sagittal** and **short-axis** fat-suppressed images as a **low-signal band** on the plantar aspect of the joint.

- Its firm **distal attachment** is onto the **base of the proximal phalanx**; the proximal attachment to the metatarsal neck is looser.
- Tears classically occur at the **distal (phalangeal) insertion** and show fluid/high signal undermining the attachment, often with adjacent MTP synovitis.

LANDMARK Collateral ligaments of the MTP joint

The **collateral ligaments** run from the metatarsal head epicondyle to the proximal phalanx base (collateral proper) and to the plantar plate (accessory collateral). Find them on **short-axis** flanking the joint medially and laterally.

- Loss of a collateral plus plantar plate tear underlies the **crossover / drifting second toe** deformity.

PITFALL Magic-angle on the plantar plate

Normal plantar plate and adjacent flexor tendon can show intermediate signal on **short-TE (PD/T1)** sequences from **magic-angle** effect. Confirm a true tear on **fluid-sensitive (T2/STIR)** images — real tears stay bright and show fluid cleft or distal detachment.

Intermetatarsal Space — Morton Neuroma and Bursa

HOW TO FIND IT Morton neuroma

Look in the **intermetatarsal space**, most often the **3rd (third/fourth) web space** and next the **2nd**, sitting **plantar to the deep transverse intermetatarsal ligament**.

- On **short-axis**, it is a teardrop/dumbbell-shaped mass that is **low-to-intermediate signal on T2** (fibrosis) and enhances; it sits plantar/between the metatarsal heads.
- It is best demonstrated on **short-axis T1/T2**; the plantar bulge below the transverse ligament is the locating cue.

PITFALL Intermetatarsal bursa vs neuroma

The **intermetatarsal bursa** lies **dorsal to the deep transverse metatarsal ligament**, is **fluid-bright (high T2)**, and is often a thin crescent. A **Morton neuroma** is **plantar** to that ligament and **low-T2/fibrous**. Position relative to the **deep transverse ligament** plus signal character separates them.

- A small dorsal fluid collection up to a few millimeters can be a normal/asymptomatic bursa — don't overcall it as a neuroma.

MEASURE Size threshold for symptomatic neuroma

Symptomatic Morton neuromas are usually **>5 mm** in transverse (short-axis) diameter. Smaller lesions are common and may be incidental — correlate with focal web-space tenderness.

Great Toe Sesamoids

HOW TO FIND IT Tibial and fibular sesamoids

Find the two sesamoids on **short-axis/axial** plantar to the **first metatarsal head**, embedded in the **flexor hallucis brevis (FHB)** tendons. The **medial = tibial** sesamoid, the **lateral = fibular** sesamoid.

- The **flexor hallucis longus (FHL)** tendon runs in the groove **between** the two sesamoids — use it to confirm medial vs lateral.

PITFALL Bipartite sesamoid vs fracture

A **bipartite (or multipartite) sesamoid** has **smooth, corticated, rounded** margins at the cleft and is more common on the **tibial (medial)** side; fragments together are often larger than a single sesamoid.

- A **fracture** shows **sharp, non-corticated, matching** edges and, on MRI, **marrow edema** in the fragments (which also occurs in sesamoiditis/osteonecrosis).
- Marrow edema is the MRI tiebreaker — present in acute fracture/stress and sesamoiditis, typically absent in a quiescent bipartite variant.

Metatarsal Stress Injury

HOW TO FIND IT Stress reaction vs stress fracture

Scan the metatarsal shafts on **fat-suppressed fluid-sensitive** images. Most metatarsal stress injuries occur in the **2nd and 3rd metatarsal shafts/necks** (march fracture).

- **Stress reaction:** periosteal/marrow edema only, no fracture line.
- **Stress fracture:** edema **plus a low-signal fracture line** on T1/PD, often with periosteal reaction.

PEARL Watch the base of the 5th metatarsal

The **proximal fifth metatarsal** has distinct zones — a tuberosity avulsion vs the **metaphyseal-diaphyseal (Jones) zone** ~1.5–2 cm distal, which is the **watershed** area prone to delayed/non-union. Localize the line precisely before calling it a Jones fracture.

Plantar Fascia

HOW TO FIND IT Plantar fascia (plantar aponeurosis)

On **sagittal** and **coronal**, trace the fascia from its origin on the **medial calcaneal tuberosity** running distally toward the toes. The thick **central cord** is the dominant band, with thinner **medial and lateral bands**.

- Normal: a thin **uniformly low-signal** band, thickest near the calcaneal origin.

MEASURE Plantar fasciitis thickness

Measure thickness at/just distal to the **calcaneal origin**. Normal is roughly **≤4 mm**; **plantar fasciitis** shows thickening (often **>4 mm**) with **fusiform high signal** in and around the fascia and **perifascial edema**.

PITFALL Plantar fibromatosis vs fasciitis vs tear

Plantar fibromatosis (Ledderhose) is a **nodular** low-T1/low-to-intermediate-T2 mass arising from the fascia, typically more **distal** (mid-arch), and is **not** at the calcaneal origin. A **tear** shows fiber discontinuity and

surrounding edema/fluid, usually after a traumatic or steroid history.

Accessory Ossicles and Variants



HOW TO FIND IT Accessory navicular (types I–III)

Look at the **medial navicular** at the **posterior tibial tendon (PTT)** insertion.

- **Type I (os tibiale externum):** small round sesamoid **within** the distal PTT, separate from the navicular.
- **Type II:** larger triangular ossicle joined to the navicular by a **synchondrosis** (cartilaginous/fibrous bridge) — the symptomatic one; look for **marrow edema across the synchondrosis** on fluid-sensitive images.
- **Type III (cornuate navicular):** **fused** prominence (medial horn) of the navicular.



HOW TO FIND IT Os peroneum

Find the **os peroneum** within the **peroneus longus tendon** as it curves around the **lateral/plantar cuboid**. It is a normal sesamoid; **edema, displacement, or a gap** in the tendon suggests **painful os peroneum syndrome** or a peroneus longus tear.



PITFALL Normal marrow heterogeneity — don't overcall edema

Red (hematopoietic) marrow and normal marrow heterogeneity in the **calcaneus, cuneiforms, and metatarsal bases** can mimic pathology. True significant edema is **bright on fat-suppressed fluid-sensitive** sequences and **dark on T1**; physiologic red marrow stays **brighter than muscle/disc on T1** and lacks a discrete lesion or cortical correlate.

- Always pair a fluid-sensitive finding with the **T1** appearance and look for an anatomic explanation (stress site, adjacent ligament/tendon injury, arthropathy) before calling marrow edema pathologic.

Infection Search Pattern

- **1. Marrow/bone changes:**
 - Marrow replacement (darker than muscle/disc on T1; geographic pattern more predictive than reticular/periarticular).
 - Marrow edema (isolated finding raises concern in diabetics/high-risk).
 - Bone erosion (loss of dark T1/T2 cortical line).
- **2. Surrounding inflammatory changes:** fascia, musculature, subcutaneous fat edema/enhancement.
- **3. Susceptibility suggesting air:** in collections, soft tissues, along fascial planes, in bone.
- **4. Abnormal enhancement:**
 - Rim-enhancing collection = abscess.
 - Superficial lesion without enhancement = ulcer.

- Enhancement along fascial planes = infectious/inflammatory fasciitis.
- Non-enhancement of normal structures = ischemia.
- Other patterns of enhancement may not be specific.
- **5. Differentiating osteomyelitis from neuropathic joint:**
- Joint effusion with marrow replacement on both sides of joint.
- Fluid collections/abscesses (periprosthetic, subperiosteal, intraosseous/Brodie's abscess).
- Sinus tract connecting abnormal marrow signal to skin surface.
- Disappearance of subchondral cysts on sequential imaging.
- Progressive marrow/soft tissue changes greater than expected on sequential imaging.

Core Measurements

Structure	Normal / Threshold	Clinical Significance
Lisfranc interval	<2 mm between medial cuneiform and 2nd metatarsal base	>2 mm suggests ligament disruption; compare with contralateral
Metatarsal stress fracture risk	2nd and 3rd metatarsal shafts/necks	Most common sites; 2nd MT especially with hallux valgus transfer
Tibial sesamoid	10-12 mm long axis	Fracture if irregular margins + marrow edema
Fibular sesamoid	8-10 mm long axis	Less commonly injured; bipartite more common
Navicular body height	Loss of height = collapse	Central 1/3 is watershed zone — highest stress fracture risk

Foot MRI Tells

- **Lisfranc:** the dorsal component tears first; always check all three components (dorsal, interosseous, plantar). Interosseous is the strongest and most important.
- **Plantar plate:** look for crossover toe deformity — suggests chronic tear. Second MTP most commonly affected.
- **Morton neuroma:** between 2nd-3rd or 3rd-4th web spaces; T1 and T2 low signal mass (fibrotic). Not a true neuroma — perineural fibrosis.
- **Navicular stress fracture:** central 1/3 at highest risk (watershed vascular zone, tension side). Sagittal plane fracture on axial images.
- **Sesamoid AVN vs stress fracture:** dual sesamoids don't enhance uniformly. Bipartite sesamoid has smooth corticated margins; fracture has irregular edges with edema.
- **Freiberg disease:** 2nd metatarsal head most common. Flattening, sclerosis, and fragmentation suggest AVN.
- **Turf toe:** first MTP plantar plate/capsule injury. Grade the severity — grade 3 involves complete plantar plate tear with dorsal subluxation.

Modified Outerbridge Chondromalacia Grading - Copy-Ready

Use this for any articular surface. Replace [surface/compartment] with the specific location: patellar median ridge, medial femoral condyle weightbearing surface, anterosuperior acetabulum, radiocapitellar joint, tibiotalar dome, first MTP joint, etc.

Grade	MRI shorthand
0	Normal cartilage.
1	Signal heterogeneity/softening or swelling; surface intact.
2	Superficial fissuring/fraying/partial-thickness defect <50% cartilage depth.
3	Deep fissuring/partial-thickness defect >50% cartilage depth, not exposing bone.
4	Full-thickness cartilage loss with exposed subchondral bone, often edema/cysts.

Grade 0 - normal cartilage

F: Articular cartilage of the [surface/compartment] is preserved in thickness and signal without focal fissuring, delamination, or subchondral marrow abnormality.

I: No chondral defect of the [surface/compartment].

Grade 1 - chondral signal change / softening

F: Mild focal cartilage signal heterogeneity/softening of the [surface/compartment] with preserved cartilage thickness and intact articular surface. No subchondral marrow edema.

I: Grade 1 chondromalacia of the [surface/compartment].

Grade 2 - superficial partial-thickness chondral loss

F: Superficial chondral fissuring/fraying of the [surface/compartment] involving less than 50% of cartilage thickness. No full-thickness defect or subchondral marrow edema.

I: Grade 2 chondromalacia of the [surface/compartment].

Grade 3 - deep partial-thickness chondral loss

F: Deep partial-thickness chondral fissuring/defect of the [surface/compartment] involving greater than 50% of cartilage thickness without exposed subchondral bone. [Mild adjacent subchondral marrow edema.]

I: Grade 3 chondromalacia of the [surface/compartment].

Grade 4 - full-thickness chondral loss

F: Full-thickness cartilage loss/fissuring of the [surface/compartment] measuring [X] x [Y] mm with exposed subchondral bone and [subchondral marrow edema/cystic change]. [Small unstable chondral flap/loose body if present.]

I: Grade 4 chondromalacia/full-thickness chondral defect of the [surface/compartment].

Normal / Negative Templates

Normal foot

F: The metatarsals, phalanges, tarsal bones, and sesamoids demonstrate normal marrow signal without fracture, stress injury, or avascular necrosis. The Lisfranc ligament complex is intact with normal alignment of the tarsometatarsal joints. The plantar plates of the MTP joints are intact. No intermetatarsal mass or neuroma. The plantar fascia is normal in thickness and signal. The tendons and ligaments are intact. No joint effusion, synovitis, or soft tissue mass.

I: Normal MRI of the foot.

Useful Caveats

- **Bipartite sesamoid vs fracture:** bipartite has smooth, well-corticated margins with no marrow edema; fracture has irregular/jagged edges with adjacent marrow edema and soft tissue swelling. Bipartite is bilateral in ~25% of cases — contralateral comparison may help.
- **Accessory navicular types:** Type I is a sesamoid within the tibialis posterior tendon (os tibiale externum); Type II is connected to the navicular by a synchondrosis (most symptomatic); Type III is a cornuate navicular (fused prominence).
- **Coalition variants:** talocalcaneal (middle facet most common) and calcaneonavicular are the two most frequent. Look for irregular joint surfaces, marrow edema, and secondary signs (talar beak, ball-and-socket ankle).

Forefoot / Stress / Lisfranc

Plantar plate tear

F: [Partial/full-thickness] tear of the [2nd] MTP plantar plate at the phalangeal attachment with pericapsular edema and [dorsal subluxation/crossover deformity]. [Collateral ligament thickening/tear.]

I: [Partial/full-thickness] [2nd] MTP plantar plate tear. [Clinical correlation for crossover toe deformity.]

Morton neuroma

F: Ovoid low-to-intermediate T1 and T2 signal soft tissue mass in the [2nd-3rd/3rd-4th] intermetatarsal space measuring [X] mm in maximum transverse dimension, plantar to the intermetatarsal ligament.

[Adjacent intermetatarsal bursitis.]

I: Morton neuroma in the [2nd-3rd/3rd-4th] intermetatarsal space.

Intermetatarsal bursitis

F: T2 hyperintense fluid distension of the [2nd-3rd/3rd-4th] intermetatarsal bursa dorsal to the intermetatarsal ligament, without discrete solid soft tissue mass. [Mild surrounding soft tissue edema.]

I: Intermetatarsal bursitis of the [2nd-3rd/3rd-4th] web space without Morton neuroma.

[Consider reactive/inflammatory etiology.]

Metatarsal stress reaction

- F:** Marrow edema of the [2nd] metatarsal shaft/neck with periosteal edema but no discrete low-signal fracture line.
- I:** Stress reaction of the [2nd] metatarsal without completed fracture.

Metatarsal stress fracture

- F:** Linear low-signal fracture line through the [2nd] metatarsal [shaft/neck] with surrounding marrow and periosteal edema. [No displacement or angulation.]
- I:** Nondisplaced stress fracture of the [2nd] metatarsal.

Navicular stress fracture

- F:** Linear low-signal fracture line in the [central/dorsal] aspect of the navicular body on axial/sagittal images with surrounding marrow edema. [Partial/complete fracture extending from the dorsal cortex.] No displacement.
- I:** Navicular stress fracture. High-risk location — recommend non-weightbearing and orthopedic/sports medicine consultation.

Lisfranc ligament tear

- F:** Disruption/edema of the Lisfranc ligament complex between the medial cuneiform and second metatarsal base. The [dorsal/interosseous/plantar] component is [partially torn/completely torn]. Adjacent marrow edema in the medial cuneiform and second metatarsal base. [Diastasis of [X] mm between the medial cuneiform and second metatarsal base.] [Associated fractures of the [cuneiform/metatarsal base].]
- I:** Lisfranc ligament injury [with/without] malalignment. [Specify components involved.]

First MTP OA

- F:** First MTP osteoarthritis with cartilage loss, marginal osteophytes, subchondral cystic change/marrow edema, and small joint effusion/synovitis. [Dorsal osteophyte measuring [X] mm limiting dorsiflexion.]
- I:** First MTP osteoarthritis/hallux rigidus.

Sesamoid fracture/AVN

- F:** [Fracture]: irregular low-signal line through the [tibial/fibular] sesamoid with surrounding marrow edema and adjacent soft tissue edema. [AVN]: diffuse low T1 and heterogeneous T2 signal within the [tibial/fibular] sesamoid with [partial/absent] enhancement, fragmentation, and surrounding soft tissue edema.
- I:** [Tibial/fibular] sesamoid [fracture/avascular necrosis].
[Recommend correlation with contralateral foot to exclude bipartite variant if fracture is suspected.]

Accessory navicular (Type II) syndrome

- F:** Type II accessory navicular with a well-corticated ossicle connected to the navicular tuberosity by a synchondrosis. Marrow edema at the synchondrosis interface and within the accessory ossicle. Surrounding soft tissue edema and [tibialis posterior tendinosis/partial tear at the insertion]. [Small joint effusion.]
- I:** Symptomatic Type II accessory navicular with synchondrosis stress reaction and [tibialis posterior tendinopathy].

Tarsal coalition

- F:** [Fibrous/cartilaginous/osseous] coalition at the [talocalcaneal middle facet/calcaeonavicular] joint with irregular articular surfaces, [marrow edema at the coalition site], and [narrowing of the posterior subtalar joint]. Secondary signs include [talar beaking/ball-and-socket ankle joint/hindfoot valgus].

I: [Talocalcaneal/calcaneonavicular] coalition, [fibrous/cartilaginous/osseous] type. [Secondary degenerative changes.]

Freiberg disease

F: Flattening and irregularity of the [2nd/3rd] metatarsal head with subchondral sclerosis, fragmentation, and surrounding marrow edema. [Overlying cartilage loss and joint effusion.] [Loose bodies within the MTP joint.]

I: Freiberg disease (metatarsal head avascular necrosis) of the [2nd/3rd] metatarsal. [Smillie stage [I-V]].

Turf toe

F:

[Grade 1: periligamentous edema surrounding the first MTP plantar plate/capsuloligamentous complex without discrete tear.]

[Grade 2: partial-thickness tear of the plantar plate at the [proximal/distal] attachment with surrounding edema and small joint effusion.] [Grade 3: complete tear of the plantar plate with

[proximal retraction/dorsal subluxation of the proximal phalanx] and [sesamoid diastasis/retraction].]

I: Turf toe injury, grade [1/2/3]. [First MTP plantar plate/capsuloligamentous tear.]

Midfoot osteoarthritis

F: Degenerative changes at the [tarsometatarsal/naviculocuneiform/intercuneiform] joints with cartilage loss, subchondral sclerosis, marginal osteophytes, and [dorsal osteophyte formation].

[Subchondral marrow edema suggesting active degenerative change.] [Small joint effusion.]

I: Midfoot osteoarthritis involving the [specify joints]. [Dorsal osteophyte impingement if symptomatic.]

Calcaneal stress fracture

F: Linear low-signal fracture line in the [posterior/anterior] calcaneal tuberosity perpendicular to the primary trabecular axis with surrounding marrow edema and periosteal edema along the calcaneal cortex. No displacement, cortical step-off, or extension to the subtalar joint surface.

[Adjacent soft tissue edema along the Achilles insertion/plantar fascia origin.]

I: Nondisplaced calcaneal stress fracture.

[Recommend clinical correlation, activity modification, and follow-up imaging if symptoms persist.]

Plantar fibromatosis (Ledderhose disease)

F: Fusiform/nodular thickening within the central cord of the plantar fascia measuring [X] x [Y] mm, demonstrating low-to-intermediate T1 and low-to-intermediate T2 signal. No aggressive features.

[Multiple nodules along the plantar aponeurosis.]

I: Plantar fibroma/fibromatosis (Ledderhose disease). Benign.

[Surgical excision if symptomatic and refractory to conservative measures.]

Source Notes

Built as a practical dictation aid from standard MSK MRI reporting patterns, review-level radiology references, and the existing shoulder cheat-sheet style. Use surgical correlation and local report conventions when needed.